Lecture 4
Hospital Software Systems and Digital Workflow

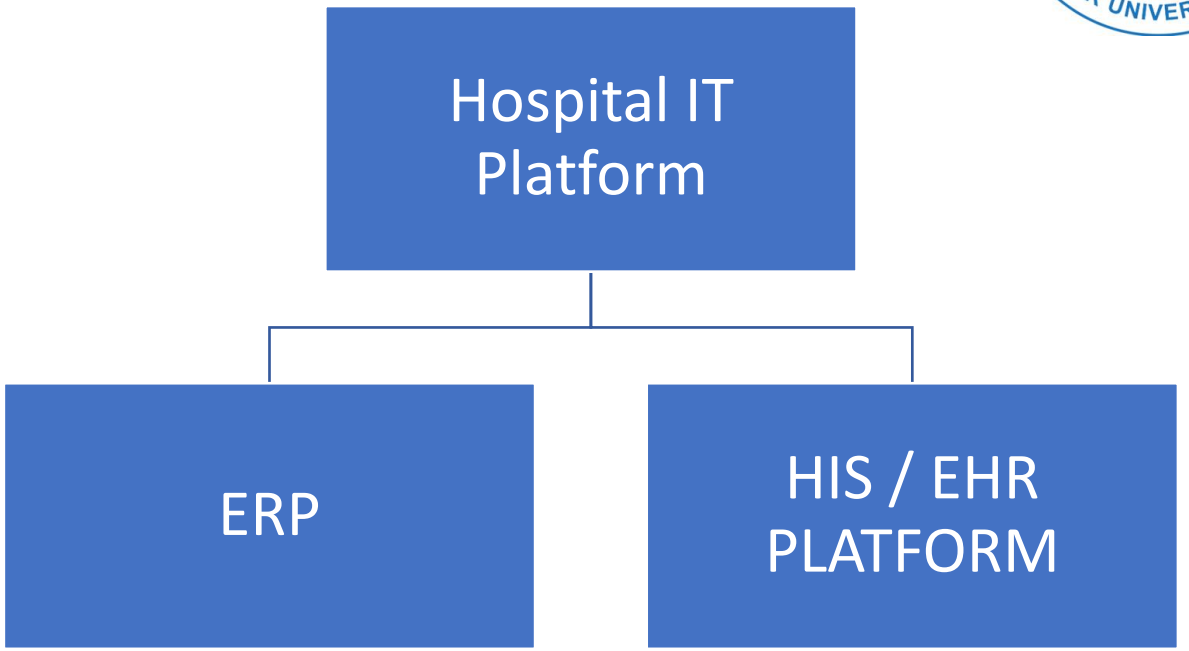
Hospital Software Overview

Overview

- Hospitals rely on multiple specialized software systems, not one single system
- HIS / EHR is the central platform coordinating clinical data
- Departmental systems (PACS, RIS, LIS) work independently and connect to HIS
- HL7 / FHIR and DICOM enable communication between systems
- ERP and CMMS support administration and engineering functions

ENTERPRISE / PLATFORM LEVEL

- ERP : Enterprise Resource Planning
- HIS : Hospital Information System
- EMR / EHR : Electronic Medical / Health Record



ERP – Enterprise Resource Planning

- Manages administrative and financial operations
- Covers HR, payroll, purchasing, inventory, accounting
- Used mainly by management and administration, not clinicians
- May share data with HIS (staff IDs, departments, costs)
- Does not handle clinical records or medical images

HIS – Hospital Information System

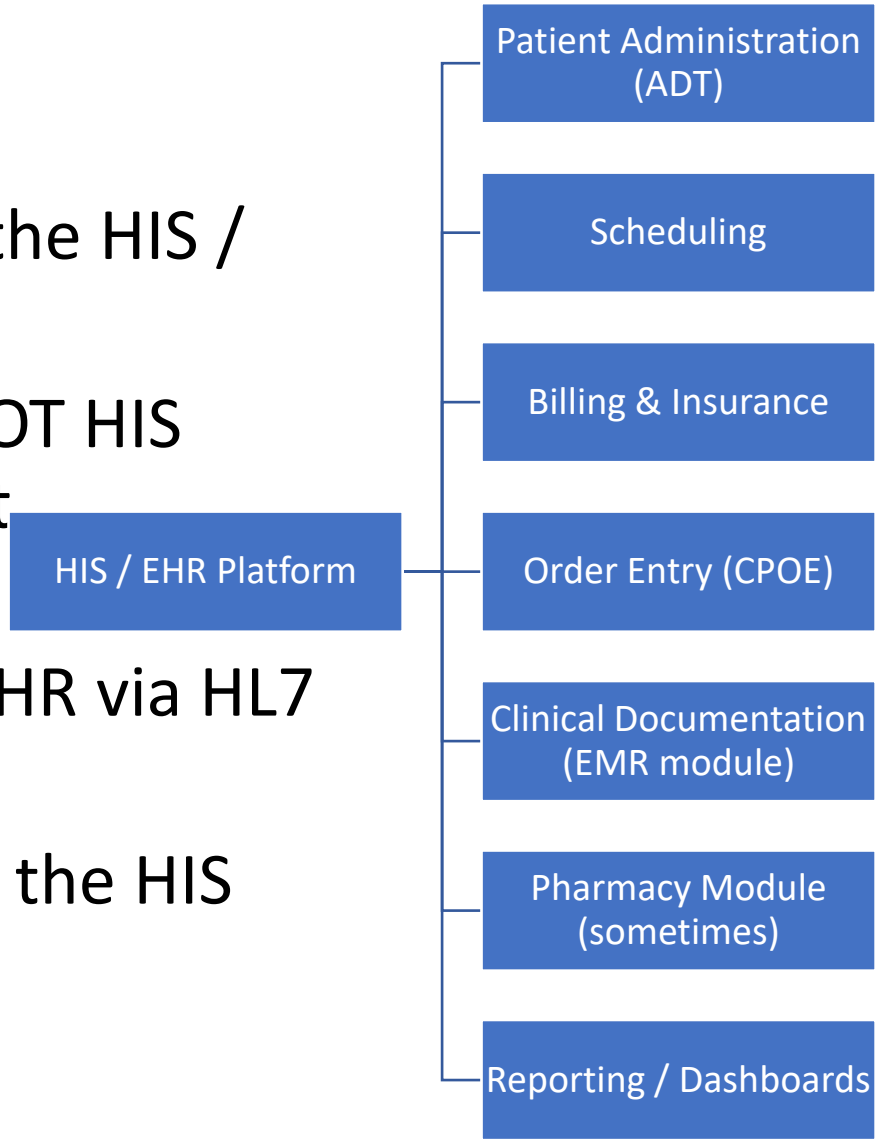
- Acts as the central coordinator of hospital operations
- Manages patient registration, appointments, billing, orders
- Connects to PACS, RIS, LIS, Pharmacy, CMMS
- Routes data between systems using HL7 / FHIR
- Considered the “brain” of hospital clinical workflow

EMR / EHR – Electronic Medical / Health Record

- Stores clinical patient data (notes, diagnoses, vitals, reports)Used by physicians, nurses, and clinical staff
- Usually a module inside HIS / EHR platform
- Contains text-based medical information, not images
- Receives results from PACS/LIS via HIS

What the HIS / EHR Platform Contains (Modules)

- EMR is usually a module inside the HIS / EHR platform
- RIS, PACS, LIS, and CMMS are NOT HIS modules , They are independent departmental systems
- These systems connect to HIS/EHR via HL7 and DICOM
- They are never contained inside the HIS



Patient Administration (ADT)

- Manages patient registration, admission, transfer, discharge
- Creates the unique patient ID used by all systems
- First step in almost every clinical workflow
- Core source of demographic data for all departments

Scheduling

- Manages appointments, clinics, rooms, and resources
- Used for outpatient visits, procedures, and follow-ups
- Sends schedule information to departments (e.g., Radiology)
- Reduces conflicts and waiting time

Billing & Insurance

- Handles charges, invoices, and insurance claims
- Links clinical activities to financial records
- Integrates with ERP and insurance providers
- Critical for hospital revenue cycle

Order Entry (CPOE)

- Physicians place orders (lab tests, imaging, medications)
- Orders are sent electronically to LIS, RIS, Pharmacy
- Reduces errors compared to paper-based orders
- Uses HL7 / FHIR for communication

Clinical Documentation (EMR module)

- Stores clinical notes, diagnoses, vitals, reports
- Used by doctors and nurses
- Text-based medical record (not images)
- Central part of patient medical history

Pharmacy Module (sometimes)

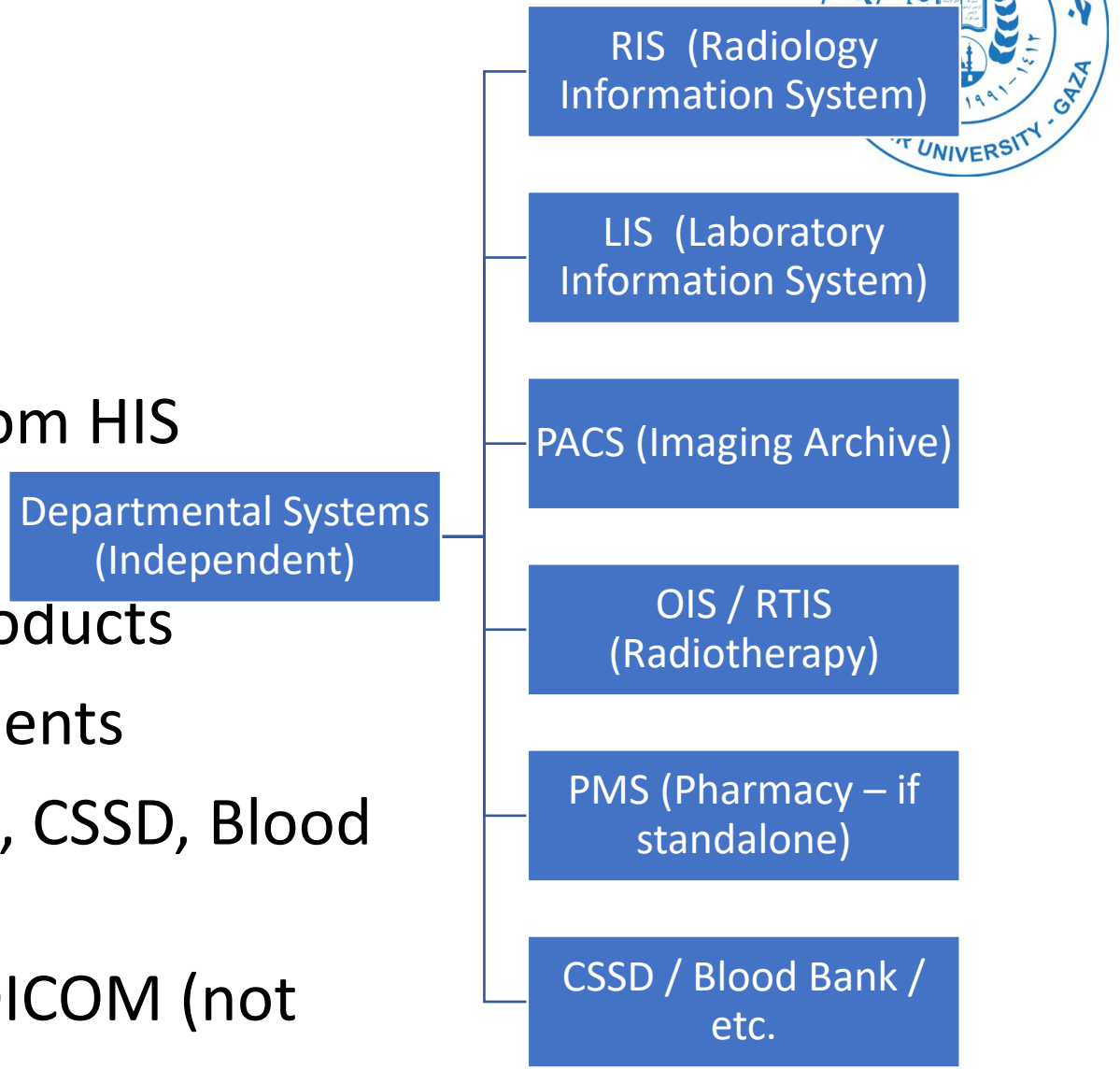
- Manages medication orders and dispensing
- Performs basic drug checks (dose, interaction)
- Often integrated or replaced by a standalone pharmacy system
- Depends on hospital size and complexity

Reporting / Dashboards

- Provides clinical, operational, and management reports
- Displays KPIs, workload, and performance indicators
- Used by clinicians and hospital management
- Supports decision-making and audits

Departmental Systems (Independent Systems)

- Not contained inside HIS
- Only connected to HIS
- Receive patient and order data from HIS
- Send results back to EMR via HIS
- Separate, standalone software products
- Used by specific hospital departments
- Examples: RIS, LIS, PACS, OIS/RTIS, CSSD, Blood Bank
- Integrated with HIS using HL7 & DICOM (not inside HIS)



RIS – Radiology Information System

- Manages radiology workflow (scheduling, reporting, status)
- Receives imaging orders from HIS
- Sends reports back to EMR via HIS
- Does not store images

LIS – Laboratory Information System

- Manages lab tests, samples, and results
- Receives lab orders from HIS
- Sends test results back to EMR
- Interfaces with analyzers and automation systems

PACS – Picture Archiving and Communication System

- Stores, retrieves, and displays medical images
- Used by radiologists and clinicians
- Communicates using DICOM
- Does not manage scheduling or reports

OIS / RTIS – Oncology / Radiotherapy Information System

- Manages radiotherapy planning and treatment workflow
- Handles treatment schedules and dose records
- Interfaces with LINAC, TPS, and imaging systems
- Sends treatment summaries to HIS/EMR

PMS – Pharmacy Management System

- Manages medication dispensing and inventory
- Receives prescriptions from HIS/CPOE
- Performs drug checks and stock control
- Sometimes integrated, sometimes standalone

CSSD / Blood Bank / Others

- CSSD: sterilization and instrument tracking
- Blood Bank: blood products, compatibility, traceability
- Operate independently within departments
- Exchange data with HIS for safety and documentation

Standards Overview

- HL7 (v2 / v3)
  - Old (still used) Used for structured clinical text data
  - Handles ADT, orders, results
  - Example: Patient admission message sent from HIS to LIS
- FHIR (HL7-FHIR)
  - Modern, web-based version of HL7
  - Uses REST APIs, JSON/XML
  - Example: Mobile app retrieves lab results from HIS using FHIR API

Standards Overview

- DICOM
  - Standard for medical images and radiation therapy data
  - Covers CT, MRI, X-ray, RT plans/dose/structures
  - Example: CT images sent from scanner to PACS in DICOM format4

- Research & AI Formats (Non-clinical)
  - Used outside routine hospital systems
  - Include NIfTI, PNG, CSV
  - Example: DICOM CT converted to NIfTI for AI segmentation training

Key Concept

- Standards connect medical systems
- They are NOT software or containers
- Used to enable interoperability between systems

HL7 (v2 / v3) – Clinical Messages

- **Type:** Text-based messages
  Used for: ADT, orders, results
- **Example HL7 (v2) message:**

```
MSH|^~\&|HIS|Hospital|LIS|Lab|20251216||ORU^R01|12345|P|2.3
PID|1||789456||Ali^Ahmad||19900101|M
OBR|1||4567|CBC
OBX|1|NM|Hb||13.5|g/dL
```

- *Meaning:* Patient admitted, lab test ordered, result sent back.

FHIR – Modern HL7 (API / Web-Based)

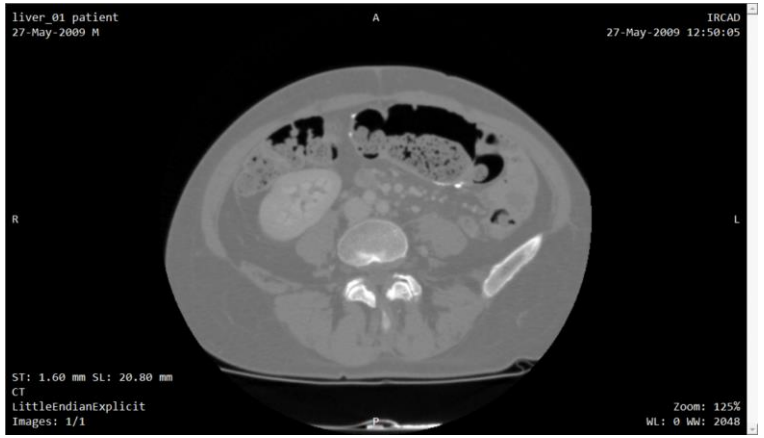
- **Type:** JSON / XML over REST API
  Used for: Apps, dashboards, cloud systems
- **Example FHIR (JSON):**

```
{
  "resourceType": "Patient",
  "id": "789456",
  "name": [{
    "family": "Ahmad",
    "given": ["Ali"]
  }],
  "gender": "male",
  "birthDate": "1990-01-01"
}
```

- *Meaning:* Mobile or web app requesting patie

DICOM – Medical Imaging & RT Data

- **Type:** Binary image + metadata
  Used for: Images, RT plans, dose, structures
- **Example DICOM metadata fields:**



- **DICOM = Image + Metadata in the same file**
- *Meaning:* CT scan stored and sent to PACS.

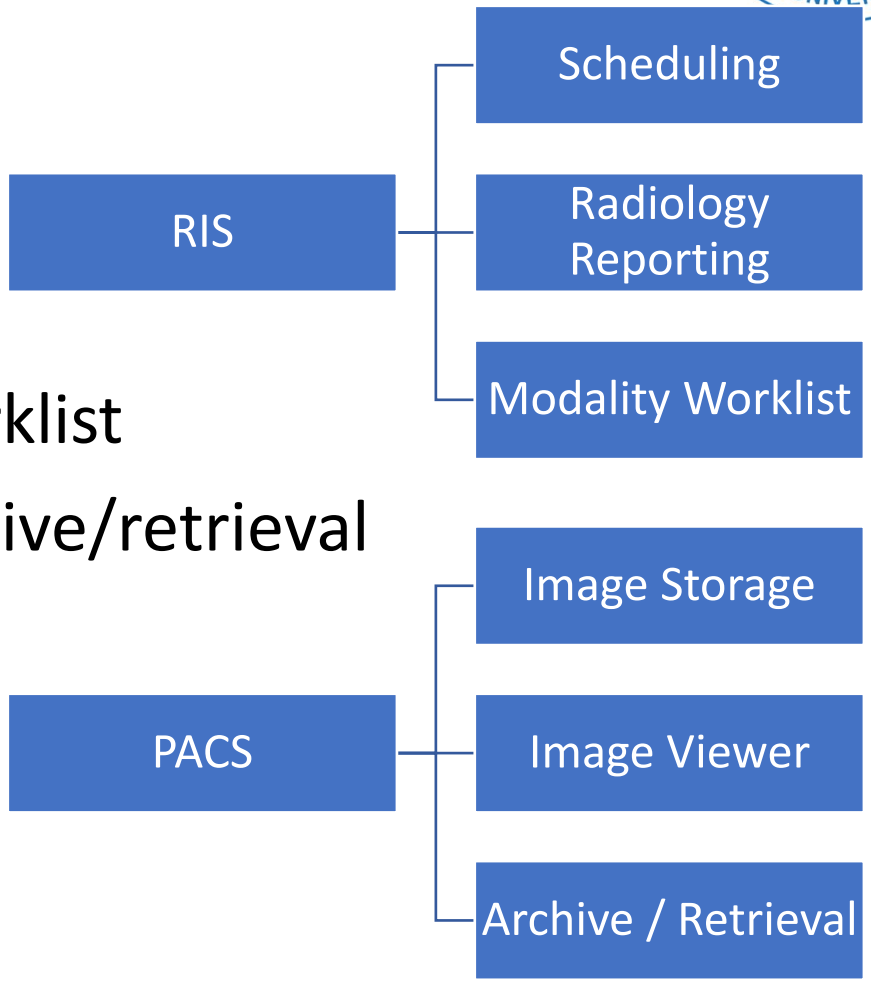
```
DICOM FILE CONTENT
_____________________________________________


METADATA (DICOM HEADER)
Patient ID / Name : liver_01 patient
Patient Sex       : Male (M)
Study Date        : 27-May-2009
Study Time        : 12:50:05
Institution       : IRCAD
Modality          : CT
Slice Thickness   : 1.60 mm
Slice Location    : 20.80 mm
Orientation       : Anterior (A)
Laterality        : Right / Left (R / L)
Transfer Syntax   : Explicit VR Little Endian

IMAGE DATA (PIXEL DATA)
Axial CT image slice
Grayscale pixel matrix
Pixel values in Hounsfield Units (HU)
```

Radiology Departmental Systems: System Relationship

- RIS does NOT contain PACS
- PACS does NOT contain RIS
- They are siblings, not parent/child
- RIS: scheduling, reporting, modality worklist
- PACS: image storage, image viewer, archive/retrieval
- RIS = workflow (text), PACS = images



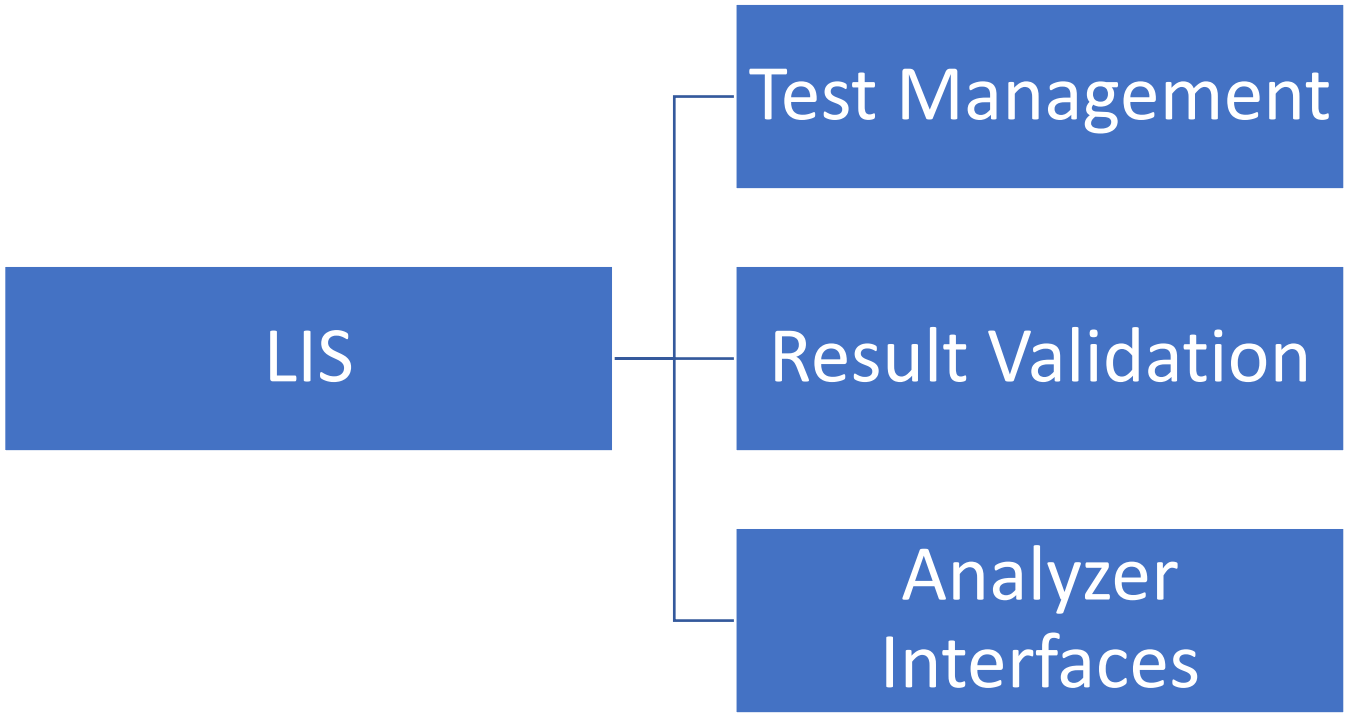
PACS – Picture Archiving and Communication System

- Image storage
- Image viewing
- Archive and retrieval
- Focus: medical images

RIS – Radiology Information System

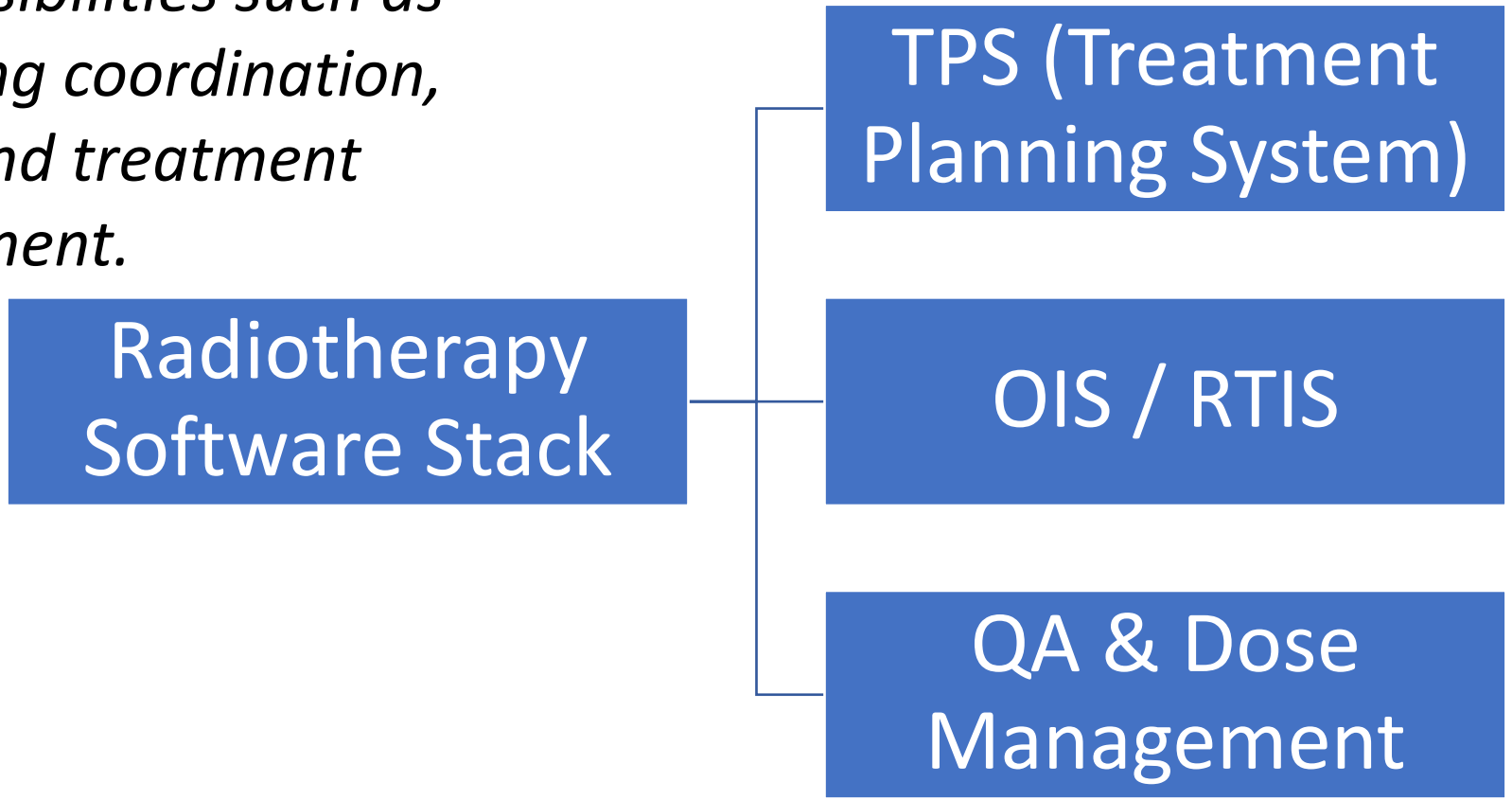
- Scheduling of imaging examinations
- Radiology reporting (text-based)
- Modality worklist management
- Focus: workflow and clinical text

Laboratory


LIS
Test Management
Result Validation
Analyzer Interfaces

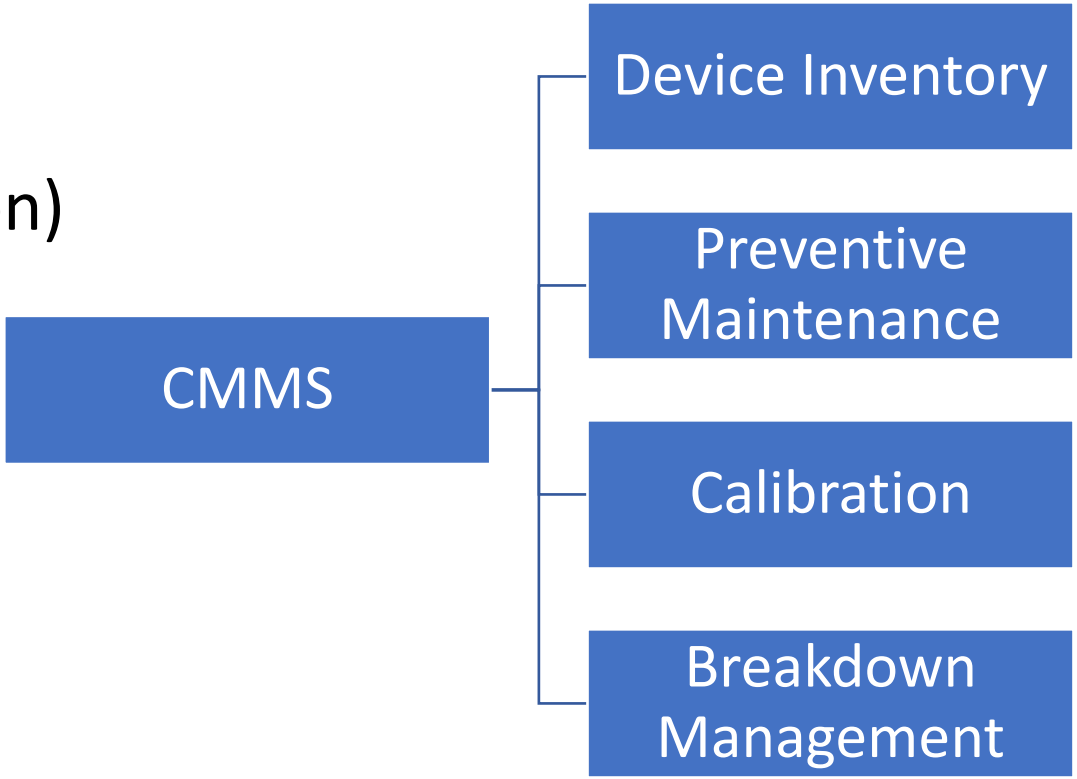
Radiotherapy

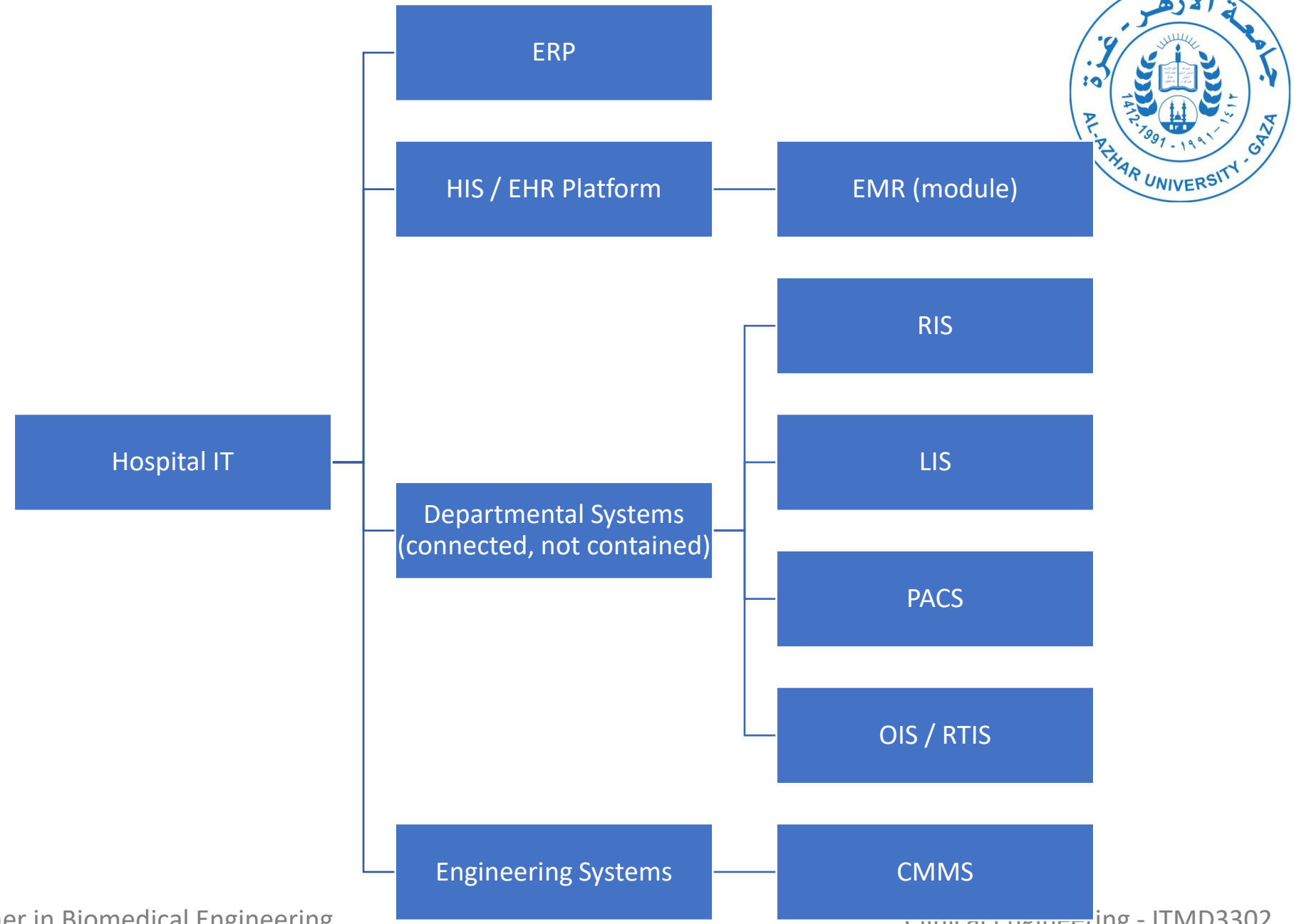
**OIS / RTIS plays a role similar to RIS, but in radiation therapy, with additional responsibilities such as treatment planning coordination, dose recording, and treatment delivery management.*



Engineering System (Separate World)

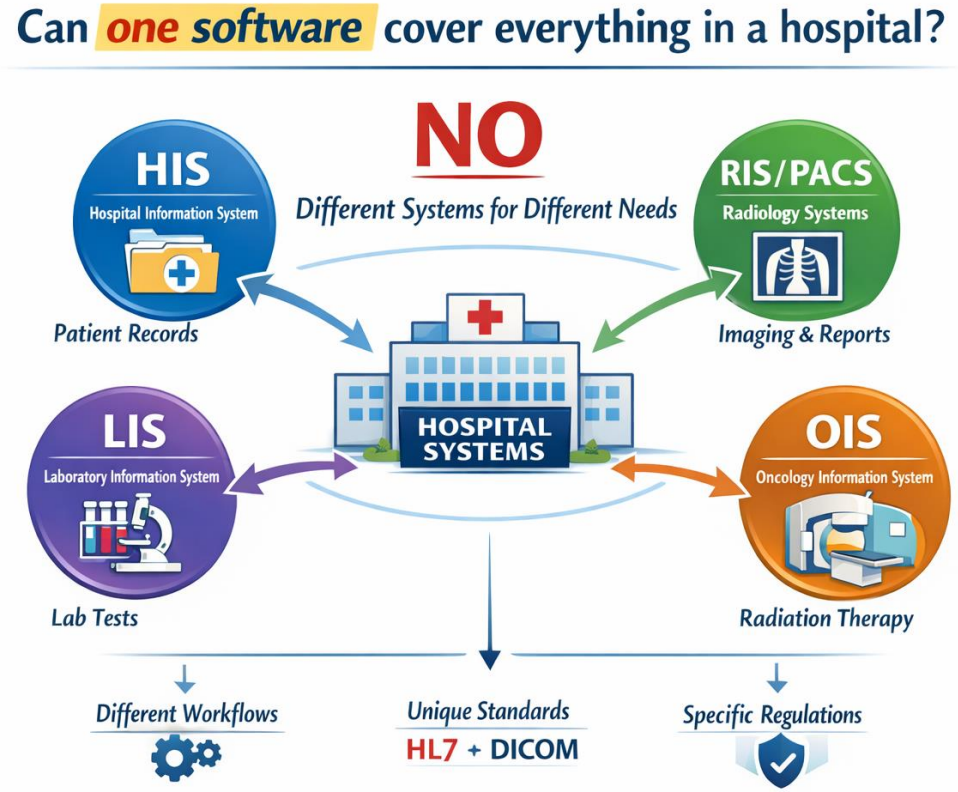
- CMMS is NOT inside HIS
- CMMS is NOT inside ERP
- It only **syncs data** (assets, location)




Hospital IT
ERP
HIS / EHR Platform
EMR (module)
Departmental Systems (connected, not contained)
RIS
LIS
PACS
OIS / RTIS
Engineering Systems
CMMS

Do we have a single software that covers everything?

- Shortly : No!
- Because hospital systems are highly specialized. Different clinical tasks (HIS, RIS, PACS, LIS, OIS) require different workflows, standards, and regulations, so they must work together, not as one system.



Think with me !? 🤔

- Why can’t one single hospital software safely replace HIS, PACS, RIS, LIS, ERP, and CMMS all together?
- If PACS stores medical images, why does the hospital still need RIS and HIS?
- HL7/FHIR and DICOM are standards, not software , what problems would hospitals face if these standards did not exist?


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Thank you